IEEE-CIS Fraud Detection

This is a jupyter notebook that is used to train and evaluate different models for the Fraud detection dataset: https://www.kaggle.com/competitions/ieee-fraud-detection/code?competitionId=14242&sortBy=voteCount&excludeNonAccessedDatasources=true

Required dependencies:

```
[
    "pandas",
    "numpy",
    "scikit-learn",
    "torch",
    "matplotlib",
    "seaborn",
    "jupyterlab",
    "xgboost"
]
```

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import roc_auc_score, classification_report, confusion_
import warnings
import gc
import os
from scipy import stats

warnings.filterwarnings('ignore')
pd.set_option('display.max_columns', 100)

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

Load the dataset

In [2]:
```
# Define data path
DATA_PATH = '~/fraud_det/dataset/'

# Load transaction data
print("Loading transaction data...")
train_transaction = pd.read_csv(os.path.join(DATA_PATH, 'train_transaction.c
test_transaction = pd.read_csv(os.path.join(DATA_PATH, 'test_transaction.csv
```

```
# Load identity data
print("Loading identity data...")
train_identity = pd.read_csv(os.path.join(DATA_PATH, 'train_identity.csv'))
test_identity = pd.read_csv(os.path.join(DATA_PATH, 'test_identity.csv'))

print(f"\nTrain Transaction Shape: {train_transaction.shape}")
print(f"Test Transaction Shape: {test_transaction.shape}")
print(f"Train Identity Shape: {train_identity.shape}")
print(f"Test Identity Shape: {test_identity.shape}")
```

```
Loading transaction data...
Loading identity data...

Train Transaction Shape: (590540, 394)
Test Transaction Shape: (506691, 393)
Train Identity Shape: (144233, 41)
Test Identity Shape: (141907, 41)
```

Merge fragmented tables

In [3]:
```
# Merge transaction and identity data on TransactionID
train = train_transaction.merge(train_identity, on='TransactionID', how='lef
test = test_transaction.merge(test_identity, on='TransactionID', how='left')

print(f"Train Shape after merge: {train.shape}")
print(f"Test Shape after merge: {test.shape}")

# Free memory
del train_transaction, test_transaction, train_identity, test_identity
gc.collect()
```

```
Train Shape after merge: (590540, 434)
Test Shape after merge: (506691, 433)
```

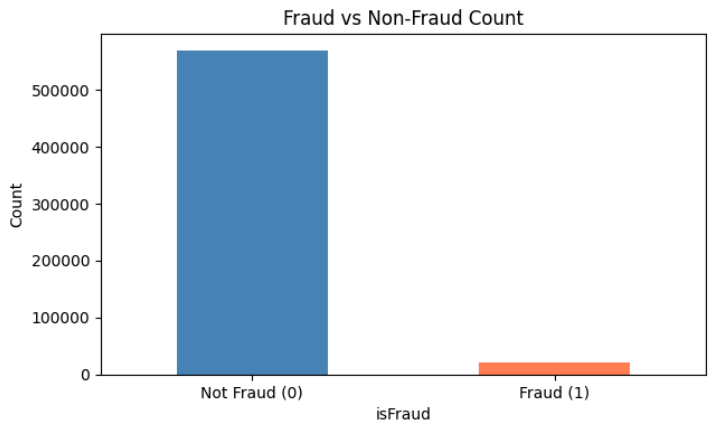
Out[3]: 0

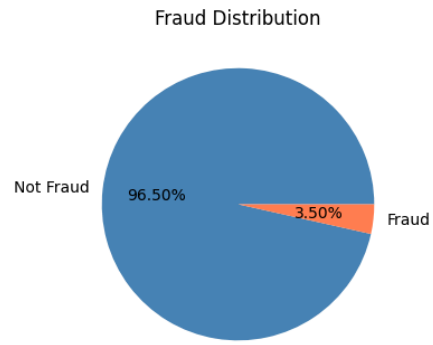
Fraud distribution in the dataset

In [4]:
```
print("Target Distribution:")
print(train['isFraud'].value_counts())
print(f"\nFraud Rate: {train['isFraud'].mean() * 100:.2f}%")

# Visualize target distribution
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

train['isFraud'].value_counts().plot(kind='bar', ax=axes[0], color=['steelbl
axes[0].set_title('Fraud vs Non-Fraud Count')
axes[0].set_xlabel('isFraud')
axes[0].set_ylabel('Count')
axes[0].set_xticklabels(['Not Fraud (0)', 'Fraud (1)'], rotation=0)

train['isFraud'].value_counts().plot(kind='pie', ax=axes[1], autopct='%1.2f%
                                     colors=['steelblue', 'coral'], labels=[
axes[1].set_title('Fraud Distribution')
```

```
axes[1].set_ylabel('')

plt.tight_layout()
plt.show()

# Column categories
print("\nColumn Categories:")
print(f"- C columns (counting): {[c for c in train.columns if c.startswith('
print(f"- D columns (timedelta): {[c for c in train.columns if c.startswith(
print(f"- M columns (match): {[c for c in train.columns if c.startswith('M')
print(f"- V columns (Vesta features): {len([c for c in train.columns if c.st
print(f"- id columns: {[c for c in train.columns if c.startswith('id_')]}")
```

```
Target Distribution:
isFraud
0    569877
1     20663
Name: count, dtype: int64

Fraud Rate: 3.50%
```






```
Column Categories:
- C columns (counting): ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7', 'C8', 'C
9', 'C10', 'C11', 'C12', 'C13', 'C14']
- D columns (timedelta): ['D1', 'D2', 'D3', 'D4', 'D5', 'D6', 'D7', 'D8', 'D
9', 'D10', 'D11', 'D12', 'D13', 'D14', 'D15', 'DeviceType', 'DeviceInfo']
- M columns (match): ['M1', 'M2', 'M3', 'M4', 'M5', 'M6', 'M7', 'M8', 'M9']
- V columns (Vesta features): 339 columns
- id columns: ['id_01', 'id_02', 'id_03', 'id_04', 'id_05', 'id_06', 'id_0
7', 'id_08', 'id_09', 'id_10', 'id_11', 'id_12', 'id_13', 'id_14', 'id_15',
'id_16', 'id_17', 'id_18', 'id_19', 'id_20', 'id_21', 'id_22', 'id_23', 'id_
24', 'id_25', 'id_26', 'id_27', 'id_28', 'id_29', 'id_30', 'id_31', 'id_32',
'id_33', 'id_34', 'id_35', 'id_36', 'id_37', 'id_38']
```

Reducing Memory usage

In [5]:
```
def reduce_mem_usage(df, verbose=True):
    """Reduce memory usage by converting data types (safe version)"""
    start_mem = df.memory_usage(deep=True).sum() / 1024**2

    for col in df.columns:
        col_type = df[col].dtype
```

```
        # Skip non-numeric columns safely
        if not pd.api.types.is_numeric_dtype(col_type):
            continue

        # Coerce to numeric to avoid string contamination
        col_data = pd.to_numeric(df[col], errors="coerce")

        if col_data.isna().all():
            # Column is completely unusable for numeric downcasting
            continue

        c_min = col_data.min()
        c_max = col_data.max()

        # Integer types
        if pd.api.types.is_integer_dtype(col_type):
            if c_min >= np.iinfo(np.int8).min and c_max <= np.iinfo(np.int8)
                df[col] = col_data.astype(np.int8)
            elif c_min >= np.iinfo(np.int16).min and c_max <= np.iinfo(np.ir
                df[col] = col_data.astype(np.int16)
            elif c_min >= np.iinfo(np.int32).min and c_max <= np.iinfo(np.ir
                df[col] = col_data.astype(np.int32)
            else:
                df[col] = col_data.astype(np.int64)

        # Float types
        else:
            if c_min >= np.finfo(np.float32).min and c_max <= np.finfo(np.fl
                df[col] = col_data.astype(np.float32)
            else:
                df[col] = col_data.astype(np.float64)

    end_mem = df.memory_usage(deep=True).sum() / 1024**2
    if verbose:
        print(
            f"Memory usage: {start_mem:.2f} MB -> {end_mem:.2f} MB "
            f"({100 * (start_mem - end_mem) / start_mem:.1f}% reduction)"
        )

    return df
```

Feature Engineering and data transformation

Some of the features have been referenced from kaggle code for better optimization: https://www.kaggle.com/competitions/ieee-fraud-detection/code?competitionId=14242&sortBy=voteCount&excludeNonAccessedDatasources=true

In [6]:
```
def create_transaction_amount_features(df):
    """Create features based on transaction amount"""
    print("Creating transaction amount features...")

    # Log transformation
    df['TransactionAmt_Log'] = np.log1p(df['TransactionAmt'])
```

```
    # Decimal part (cents) - often fraudsters use round numbers
    df['TransactionAmt_decimal'] = ((df['TransactionAmt'] - df['TransactionA

    # Is round amount (no cents)
    df['TransactionAmt_is_round'] = (df['TransactionAmt'] == df['Transactior

    # Amount bins
    df['TransactionAmt_bin'] = pd.cut(df['TransactionAmt'],
                                      bins=[0, 50, 100, 200, 500, 1000, 5000,
                                      labels=[0, 1, 2, 3, 4, 5, 6, 7]).astype

    # Cents value
    df['TransactionAmt_cents'] = (df['TransactionAmt'] * 100 % 100).astype(i

    # Common fraud amounts (ends in .00, .99, .95)
    df['TransactionAmt_ends_00'] = (df['TransactionAmt_cents'] == 0).astype(
    df['TransactionAmt_ends_99'] = (df['TransactionAmt_cents'] == 99).astype
    df['TransactionAmt_ends_95'] = (df['TransactionAmt_cents'] == 95).astype

    return df

# Apply
train = create_transaction_amount_features(train)
test = create_transaction_amount_features(test)
```

```
Creating transaction amount features...
Creating transaction amount features...
```

In [7]:

```
def create_time_features(df):
    """Create time-based features from TransactionDT"""
    print("Creating time features...")

    # TransactionDT is seconds from a reference point
    # Convert to interpretable time units

    # Hour of day (0-23)
    df['Transaction_hour'] = np.floor(df['TransactionDT'] / 3600) % 24

    # Day of week (0-6)
    df['Transaction_dow'] = np.floor(df['TransactionDT'] / (3600 * 24)) % 7

    # Day of month approximation
    df['Transaction_dom'] = np.floor(df['TransactionDT'] / (3600 * 24)) % 30

    # Week number
    df['Transaction_week'] = np.floor(df['TransactionDT'] / (3600 * 24 * 7))

    # Is weekend
    df['Transaction_is_weekend'] = (df['Transaction_dow'] >= 5).astype(int)

    # Time of day categories
    df['Transaction_time_of_day'] = pd.cut(df['Transaction_hour'],
                                           bins=[-1, 6, 12, 18, 24],
                                           labels=[0, 1, 2, 3]).astype(floa
```

```
    # Is night (00:00 - 06:00) - higher fraud risk
    df['Transaction_is_night'] = ((df['Transaction_hour'] >= 0) & (df['Trans

    # Is business hours (09:00 - 17:00)
    df['Transaction_is_business_hours'] = ((df['Transaction_hour'] >= 9) & (

    return df

# Apply
train = create_time_features(train)
test = create_time_features(test)
```

```
Creating time features...
Creating time features...
```

In [8]:

```
def create_card_features(df):
    """Create features based on card information"""
    print("Creating card features...")

    # Card combination features
    card_cols = ['card1', 'card2', 'card3', 'card4', 'card5', 'card6']

    for col in card_cols:
        if col in df.columns:
            # Fill NaN for string operations
            df[col] = df[col].fillna(-999)

    # Card1 + Card2 combination
    if 'card1' in df.columns and 'card2' in df.columns:
        df['card1_card2'] = df['card1'].astype(str) + '_' + df['card2'].asty

    # Card type combinations
    if 'card4' in df.columns and 'card6' in df.columns:
        df['card4_card6'] = df['card4'].astype(str) + '_' + df['card6'].asty

    # Card + Address combinations
    if 'card1' in df.columns and 'addr1' in df.columns:
        df['card1_addr1'] = df['card1'].astype(str) + '_' + df['addr1'].asty

    if 'card1' in df.columns and 'addr2' in df.columns:
        df['card1_addr2'] = df['card1'].astype(str) + '_' + df['addr2'].asty

    if 'card2' in df.columns and 'addr1' in df.columns:
        df['card2_addr1'] = df['card2'].astype(str) + '_' + df['addr1'].asty

    # Card + Product combinations
    if 'card1' in df.columns and 'ProductCD' in df.columns:
        df['card1_ProductCD'] = df['card1'].astype(str) + '_' + df['ProductC

    if 'card2' in df.columns and 'ProductCD' in df.columns:
        df['card2_ProductCD'] = df['card2'].astype(str) + '_' + df['ProductC

    # Card + Time combinations
    if 'card1' in df.columns and 'Transaction_hour' in df.columns:
        df['card1_hour'] = df['card1'].astype(str) + '_' + df['Transaction_h
```

```
    if 'card1' in df.columns and 'Transaction_dow' in df.columns:
        df['card1_dow'] = df['card1'].astype(str) + '_' + df['Transaction_dc

    return df

# Apply
train = create_card_features(train)
test = create_card_features(test)
```

```
Creating card features...
Creating card features...
```

In [9]:

```
def create_email_features(df):
    """Create features based on email domains"""
    print("Creating email features...")

    # P_emaildomain features
    if 'P_emaildomain' in df.columns:
        # Extract domain prefix
        df['P_emaildomain_prefix'] = df['P_emaildomain'].apply(
            lambda x: str(x).split('.')[0] if pd.notna(x) else 'unknown'
        )

        # Extract domain suffix (com, net, org, etc.)
        df['P_emaildomain_suffix'] = df['P_emaildomain'].apply(
            lambda x: str(x).split('.')[-1] if pd.notna(x) and '.' in str(x)
        )

        # Is common email provider
        common_providers = ['gmail', 'yahoo', 'hotmail', 'outlook', 'aol', '
        df['P_email_is_common'] = df['P_emaildomain_prefix'].apply(
            lambda x: 1 if x.lower() in common_providers else 0
        )

        # Is business email (not common provider)
        df['P_email_is_business'] = (df['P_email_is_common'] == 0).astype(ir

        # Email domain length
        df['P_emaildomain_len'] = df['P_emaildomain'].apply(
            lambda x: len(str(x)) if pd.notna(x) else 0
        )

    # R_emaildomain features
    if 'R_emaildomain' in df.columns:
        df['R_emaildomain_prefix'] = df['R_emaildomain'].apply(
            lambda x: str(x).split('.')[0] if pd.notna(x) else 'unknown'
        )

        df['R_emaildomain_suffix'] = df['R_emaildomain'].apply(
            lambda x: str(x).split('.')[-1] if pd.notna(x) and '.' in str(x)
        )

        df['R_email_is_common'] = df['R_emaildomain_prefix'].apply(
            lambda x: 1 if x.lower() in ['gmail', 'yahoo', 'hotmail', 'outlc
        )
```

```
    # Email match features
    if 'P_emaildomain' in df.columns and 'R_emaildomain' in df.columns:
        # Same email domain
        df['email_domain_match'] = (df['P_emaildomain'] == df['R_emaildomair

        # Same email prefix
        df['email_prefix_match'] = (df['P_emaildomain_prefix'] == df['R_emai

        # Both emails missing
        df['both_emails_missing'] = (df['P_emaildomain'].isna() & df['R_emai

        # Only P email present
        df['only_P_email'] = (df['P_emaildomain'].notna() & df['R_emaildomai

        # Only R email present
        df['only_R_email'] = (df['P_emaildomain'].isna() & df['R_emaildomair

    return df

# Apply
train = create_email_features(train)
test = create_email_features(test)
```

```
Creating email features...
Creating email features...
```

In [10]:

```
def create_device_features(df):
    """Create features based on device information"""
    print("Creating device features...")

    # DeviceType
    if 'DeviceType' in df.columns:
        df['DeviceType_is_mobile'] = (df['DeviceType'] == 'mobile').astype(i
        df['DeviceType_is_desktop'] = (df['DeviceType'] == 'desktop').astype

    # DeviceInfo
    if 'DeviceInfo' in df.columns:
        # Extract device brand
        df['Device_brand'] = df['DeviceInfo'].apply(
            lambda x: str(x).split('/')[0].split()[0] if pd.notna(x) else 'u
        )

        # Device info length (complexity indicator)
        df['DeviceInfo_len'] = df['DeviceInfo'].apply(
            lambda x: len(str(x)) if pd.notna(x) else 0
        )

        # Is specific device types
        df['Device_is_Samsung'] = df['DeviceInfo'].apply(
            lambda x: 1 if pd.notna(x) and 'samsung' in str(x).lower() else
        )
        df['Device_is_iOS'] = df['DeviceInfo'].apply(
            lambda x: 1 if pd.notna(x) and ('ios' in str(x).lower() or 'iphc
        )
        df['Device_is_Windows'] = df['DeviceInfo'].apply(
            lambda x: 1 if pd.notna(x) and 'windows' in str(x).lower() else
```

```
        )

    # Browser features from id_31
    if 'id_31' in df.columns:
        df['Browser'] = df['id_31'].apply(
            lambda x: str(x).split()[0].lower() if pd.notna(x) else 'unknowr
        )

        df['Browser_is_chrome'] = df['id_31'].apply(
            lambda x: 1 if pd.notna(x) and 'chrome' in str(x).lower() else 0
        )
        df['Browser_is_safari'] = df['id_31'].apply(
            lambda x: 1 if pd.notna(x) and 'safari' in str(x).lower() else 0
        )
        df['Browser_is_firefox'] = df['id_31'].apply(
            lambda x: 1 if pd.notna(x) and 'firefox' in str(x).lower() else
        )
        df['Browser_is_edge'] = df['id_31'].apply(
            lambda x: 1 if pd.notna(x) and 'edge' in str(x).lower() else 0
        )

    # OS features from id_30
    if 'id_30' in df.columns:
        df['OS'] = df['id_30'].apply(
            lambda x: str(x).split()[0].lower() if pd.notna(x) else 'unknowr
        )

        df['OS_is_Windows'] = df['id_30'].apply(
            lambda x: 1 if pd.notna(x) and 'windows' in str(x).lower() else
        )
        df['OS_is_Mac'] = df['id_30'].apply(
            lambda x: 1 if pd.notna(x) and 'mac' in str(x).lower() else 0
        )
        df['OS_is_iOS'] = df['id_30'].apply(
            lambda x: 1 if pd.notna(x) and 'ios' in str(x).lower() else 0
        )
        df['OS_is_Android'] = df['id_30'].apply(
            lambda x: 1 if pd.notna(x) and 'android' in str(x).lower() else
        )

    # Screen resolution from id_33
    if 'id_33' in df.columns:
        df['Screen_width'] = df['id_33'].apply(
            lambda x: int(str(x).split('x')[0]) if pd.notna(x) and 'x' in st
        )
        df['Screen_height'] = df['id_33'].apply(
            lambda x: int(str(x).split('x')[1]) if pd.notna(x) and 'x' in st
        )
        df['Screen_area'] = df['Screen_width'] * df['Screen_height']
        df['Screen_aspect_ratio'] = df.apply(
            lambda row: row['Screen_width'] / row['Screen_height'] if row['S
        )

    return df

# Apply
```

```
train = create_device_features(train)
test = create_device_features(test)
```

```
Creating device features...
Creating device features...
```

In [11]:
```
def create_address_features(df):
    """Create features based on address information"""
    print("Creating address features...")

    # Address combinations
    if 'addr1' in df.columns and 'addr2' in df.columns:
        df['addr1_addr2'] = df['addr1'].astype(str) + '_' + df['addr2'].asty

    # Address + ProductCD
    if 'addr1' in df.columns and 'ProductCD' in df.columns:
        df['addr1_ProductCD'] = df['addr1'].astype(str) + '_' + df['ProductC

    # Address distance from P_emaildomain (proxy for geographic mismatch)
    if 'addr1' in df.columns:
        df['addr1_missing'] = df['addr1'].isna().astype(int)

    if 'addr2' in df.columns:
        df['addr2_missing'] = df['addr2'].isna().astype(int)

    # Both addresses missing
    if 'addr1' in df.columns and 'addr2' in df.columns:
        df['both_addr_missing'] = (df['addr1'].isna() & df['addr2'].isna()).

    # dist1 and dist2 features
    if 'dist1' in df.columns:
        df['dist1_missing'] = df['dist1'].isna().astype(int)
        df['dist1_log'] = np.log1p(df['dist1'].fillna(0))

    if 'dist2' in df.columns:
        df['dist2_missing'] = df['dist2'].isna().astype(int)
        df['dist2_log'] = np.log1p(df['dist2'].fillna(0))

    return df

# Apply
train = create_address_features(train)
test = create_address_features(test)
```

```
Creating address features...
Creating address features...
```

In [12]:
```
def create_v_features(df):
    """Create aggregation features from V columns"""
    print("Creating V-column aggregation features...")

    # Get all V columns
    v_cols = [col for col in df.columns if col.startswith('V')]

    if len(v_cols) == 0:
        return df
```

```
    # Group V columns by their correlation patterns (based on EDA from compe
    v_groups = {
        'v1': ['V1', 'V2', 'V3', 'V4', 'V5', 'V6', 'V7', 'V8', 'V9', 'V10',
        'v2': ['V12', 'V13', 'V14', 'V15', 'V16', 'V17', 'V18', 'V19', 'V20'
        'v3': ['V27', 'V28', 'V29', 'V30', 'V31', 'V32', 'V33', 'V34'],
        'v4': ['V35', 'V36', 'V37', 'V38', 'V39', 'V40', 'V41', 'V42', 'V43'
        'v5': ['V53', 'V54', 'V55', 'V56', 'V57', 'V58', 'V59', 'V60', 'V61'
        'v6': ['V75', 'V76', 'V77', 'V78', 'V79', 'V80', 'V81', 'V82', 'V83'
        'v7': ['V95', 'V96', 'V97', 'V98', 'V99', 'V100', 'V101', 'V102', 'V
    }

    for group_name, group_cols in v_groups.items():
        # Filter to existing columns
        existing_cols = [col for col in group_cols if col in df.columns]

        if len(existing_cols) > 0:
            # Sum of group
            df[f'{group_name}_sum'] = df[existing_cols].sum(axis=1)

            # Mean of group
            df[f'{group_name}_mean'] = df[existing_cols].mean(axis=1)

            # Std of group
            df[f'{group_name}_std'] = df[existing_cols].std(axis=1)

            # NaN count in group
            df[f'{group_name}_nan_count'] = df[existing_cols].isna().sum(axi

    # Overall V statistics
    existing_v_cols = [col for col in v_cols if col in df.columns]
    if len(existing_v_cols) > 0:
        df['V_sum_all'] = df[existing_v_cols].sum(axis=1)
        df['V_mean_all'] = df[existing_v_cols].mean(axis=1)
        df['V_std_all'] = df[existing_v_cols].std(axis=1)
        df['V_nan_count_all'] = df[existing_v_cols].isna().sum(axis=1)
        df['V_nan_ratio'] = df['V_nan_count_all'] / len(existing_v_cols)

    return df

# Apply
train = create_v_features(train)
test = create_v_features(test)
```

```
Creating V-column aggregation features...
Creating V-column aggregation features...
```

In [13]:

```
def create_c_d_features(df):
    """Create features from C (count) and D (timedelta) columns"""
    print("Creating C and D column features...")

    # C columns (counting features)
    c_cols = [col for col in df.columns if col.startswith('C') and col[1:].i

    if len(c_cols) > 0:
        existing_c_cols = [col for col in c_cols if col in df.columns]
```

```
        # Aggregations
        df['C_sum'] = df[existing_c_cols].sum(axis=1)
        df['C_mean'] = df[existing_c_cols].mean(axis=1)
        df['C_std'] = df[existing_c_cols].std(axis=1)
        df['C_max'] = df[existing_c_cols].max(axis=1)
        df['C_min'] = df[existing_c_cols].min(axis=1)

        # Log transformations for key C columns
        for col in ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7', 'C8', 'C9', 'C
            if col in df.columns:
                df[f'{col}_log'] = np.log1p(df[col].fillna(0))

    # D columns (timedelta features)
    d_cols = [col for col in df.columns if col.startswith('D') and col[1:].i

    if len(d_cols) > 0:
        existing_d_cols = [col for col in d_cols if col in df.columns]

        # Aggregations
        df['D_sum'] = df[existing_d_cols].sum(axis=1)
        df['D_mean'] = df[existing_d_cols].mean(axis=1)
        df['D_std'] = df[existing_d_cols].std(axis=1)
        df['D_nan_count'] = df[existing_d_cols].isna().sum(axis=1)

        # D1 is often important (days since card was first used)
        if 'D1' in df.columns:
            df['D1_missing'] = df['D1'].isna().astype(int)
            df['D1_log'] = np.log1p(df['D1'].fillna(0))

        # D15 is also important
        if 'D15' in df.columns:
            df['D15_missing'] = df['D15'].isna().astype(int)
            df['D15_log'] = np.log1p(df['D15'].fillna(0))

    return df

# Apply
train = create_c_d_features(train)
test = create_c_d_features(test)
```

```
Creating C and D column features...
Creating C and D column features...
```

In [14]:

```
def create_m_features(df):
    """Create features from M (match) columns"""
    print("Creating M column features...")

    m_cols = ['M1', 'M2', 'M3', 'M4', 'M5', 'M6', 'M7', 'M8', 'M9']
    existing_m_cols = [col for col in m_cols if col in df.columns]

    if len(existing_m_cols) > 0:
        # Count of True values
        for col in existing_m_cols:
            df[f'{col}_isT'] = (df[col] == 'T').astype(int)
            df[f'{col}_isF'] = (df[col] == 'F').astype(int)
            df[f'{col}_missing'] = df[col].isna().astype(int)
```

```
        # Total True count
        t_cols = [f'{col}_isT' for col in existing_m_cols]
        df['M_true_count'] = df[t_cols].sum(axis=1)

        # Total False count
        f_cols = [f'{col}_isF' for col in existing_m_cols]
        df['M_false_count'] = df[f_cols].sum(axis=1)

        # Total missing count
        m_missing_cols = [f'{col}_missing' for col in existing_m_cols]
        df['M_missing_count'] = df[m_missing_cols].sum(axis=1)

        # Ratio of True to total non-missing
        df['M_true_ratio'] = df['M_true_count'] / (df['M_true_count'] + df['

    return df

# Apply
train = create_m_features(train)
test = create_m_features(test)
```

```
Creating M column features...
Creating M column features...
```

In [15]:

```
def create_aggregation_features(train_df, test_df):
    """Create frequency and aggregation features"""
    print("Creating aggregation features...")

    # Combine for consistent encoding
    train_df['is_train'] = 1
    test_df['is_train'] = 0
    combined = pd.concat([train_df, test_df], axis=0, ignore_index=True)

    # Columns for frequency encoding
    freq_cols = ['card1', 'card2', 'card3', 'card4', 'card5', 'card6',
                 'addr1', 'addr2', 'P_emaildomain', 'R_emaildomain',
                 'ProductCD', 'DeviceType', 'DeviceInfo']

    for col in freq_cols:
        if col in combined.columns:
            # Frequency encoding
            freq = combined[col].value_counts().to_dict()
            combined[f'{col}_freq'] = combined[col].map(freq)

    # Transaction amount aggregations by card
    agg_cols = ['card1', 'card2', 'card1_card2', 'addr1']

    for col in agg_cols:
        if col in combined.columns:
            # Mean transaction amount
            agg = combined.groupby(col)['TransactionAmt'].mean().to_dict()
            combined[f'{col}_TransactionAmt_mean'] = combined[col].map(agg)

            # Std transaction amount
            agg = combined.groupby(col)['TransactionAmt'].std().to_dict()
```

```
            combined[f'{col}_TransactionAmt_std'] = combined[col].map(agg)

            # Transaction amount deviation from mean
            combined[f'{col}_TransactionAmt_dev'] = combined['TransactionAmt

    # Split back
    train_df = combined[combined['is_train'] == 1].drop('is_train', axis=1)
    test_df = combined[combined['is_train'] == 0].drop('is_train', axis=1)

    del combined
    gc.collect()

    return train_df, test_df

# Apply
train, test = create_aggregation_features(train, test)
```

```
Creating aggregation features...
```

In [16]:

```
def create_id_features(df):
    """Create features from identity columns"""
    print("Creating ID features...")

    # id_01 to id_11 are numerical
    id_num_cols = [f'id_0{i}' for i in range(1, 10)] + ['id_10', 'id_11']
    existing_id_num = [col for col in id_num_cols if col in df.columns]

    if len(existing_id_num) > 0:
        df['id_num_nan_count'] = df[existing_id_num].isna().sum(axis=1)
        df['id_num_mean'] = df[existing_id_num].mean(axis=1)
        df['id_num_std'] = df[existing_id_num].std(axis=1)

    # id_12 to id_38 are categorical
    # Check specific important ones
    if 'id_12' in df.columns:
        df['id_12_isFound'] = (df['id_12'] == 'Found').astype(int)

    if 'id_15' in df.columns:
        df['id_15_isNew'] = (df['id_15'] == 'New').astype(int)
        df['id_15_isFound'] = (df['id_15'] == 'Found').astype(int)

    if 'id_16' in df.columns:
        df['id_16_isFound'] = (df['id_16'] == 'Found').astype(int)

    if 'id_28' in df.columns:
        df['id_28_isNew'] = (df['id_28'] == 'New').astype(int)
        df['id_28_isFound'] = (df['id_28'] == 'Found').astype(int)

    if 'id_29' in df.columns:
        df['id_29_isFound'] = (df['id_29'] == 'Found').astype(int)

    # id_34 (match status)
    if 'id_34' in df.columns:
        df['id_34_match'] = df['id_34'].apply(
            lambda x: int(str(x).split(':')[1]) if pd.notna(x) and ':' in st
        )
```

```
    # id_36 features
    if 'id_36' in df.columns:
        df['id_36_isT'] = (df['id_36'] == 'T').astype(int)

    # id_37, id_38 features
    if 'id_37' in df.columns:
        df['id_37_isT'] = (df['id_37'] == 'T').astype(int)

    if 'id_38' in df.columns:
        df['id_38_isT'] = (df['id_38'] == 'T').astype(int)

    return df

# Apply
train = create_id_features(train)
test = create_id_features(test)
```

```
Creating ID features...
Creating ID features...
```

In [17]:

```
print(f"\nFinal Train Shape: {train.shape}")
print(f"Final Test Shape: {test.shape}")

# Count new features
original_cols = 434  # Approximate original column count after merge
new_features = train.shape[1] - original_cols
print(f"\nNew features created: ~{new_features}")

# # Reduce memory
# print("\nReducing memory usage...")
# train = reduce_mem_usage(train)
# test = reduce_mem_usage(test)

# gc.collect()
```

```
Final Train Shape: (590540, 670)
Final Test Shape: (506691, 670)

New features created: ~236
```

In [18]:

```
# # Reduce memory
print("\nReducing memory usage...")
train = reduce_mem_usage(train)
test = reduce_mem_usage(test)

gc.collect()
```

```
Reducing memory usage...
Memory usage: 4266.48 MB -> 2815.72 MB (34.0% reduction)
Memory usage: 3647.35 MB -> 2429.64 MB (33.4% reduction)
```

Out[18]: 0

In [19]:

```
X = train.drop(['isFraud', 'TransactionID'], axis=1)
y = train['isFraud']
```

Simple Linear Regression

Choose only one variable thats independent for this excercise: TransactionAmt This is not gonna do well even for baseline predicitons as TransactionAmt is the small part of the whole dataset. Can become difficult for this one column to represent the whole data

In [20]:
```
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import r2_score, confusion_matrix

X_simple = X[['TransactionAmt']]
y_simple = y

X_train, X_test, y_train, y_test = train_test_split(
    X_simple, y_simple, test_size=0.2, random_state=42
)

lin_simple = LinearRegression()
lin_simple.fit(X_train, y_train)

y_pred = lin_simple.predict(X_test)

print("Intercept:", lin_simple.intercept_)
print("Coefficient:", lin_simple.coef_[0])
print("R²:", r2_score(y_test, y_pred))
```

```
Intercept: 0.033393342
Coefficient: 1.0107775e-05
R²: -8.797645568847656e-05
```

Multiple/Multivariate Linear Regression

Here we start using more of the cols from the data. We can choose cols arbitrarily or may be with a strategy here. Strategies may have holes but it is fine for exploratory analysis

Let's use these cols as a baseline. They seem obvious first choice so thats why we are choosing them:

- TransactionAmt
- card1_freq
- card1_TransactionAmt_mean
- card1_TransactionAmt_dev
- addr1_freq

In [21]:
```
features = [
    'TransactionAmt',
    'card1_freq',
    'card1_TransactionAmt_mean',
```

```
    'card1_TransactionAmt_dev',
    'addr1_freq'
]

X_multiple = X[features].fillna(0)
y_multiple = y

X_train, X_test, y_train, y_test = train_test_split(
    X_multiple, y_multiple, test_size=0.2, random_state=42
)
lin_multiple = LinearRegression()
lin_multiple.fit(X_train, y_train)

y_pred = lin_multiple.predict(X_test)
print("\nMultiple Linear Regression:")
print("Intercept:", lin_multiple.intercept_)
print("Coefficients:", dict(zip(features, lin_multiple.coef_)))
print("R²:", r2_score(y_test, y_pred))
```

```
Multiple Linear Regression:
Intercept: 0.05282118
Coefficients: {'TransactionAmt': np.float32(-2.0582505e-10), 'card1_freq': n
p.float32(-1.282535e-07), 'card1_TransactionAmt_mean': np.float32(-1.4518574
e-10), 'card1_TransactionAmt_dev': np.float32(-6.063483e-11), 'addr1_freq':
np.float32(-4.7757425e-07)}
R²: 0.0059931278228759766
```

Logistic regression

Same feature set pretty much:

- TransactionAmt
- card1_freq
- card1_TransactionAmt_mean
- card1_TransactionAmt_dev
- addr1_freq

In [22]:

```
features = [
    'TransactionAmt',
    'card1_freq',
    'card1_TransactionAmt_mean',
    'card1_TransactionAmt_dev',
    'addr1_freq'
]

X_logistic = X[features].fillna(0)
y_logistic = y

X_train, X_test, y_train, y_test = train_test_split(
    X_logistic, y_logistic, test_size=0.2, random_state=42
)
logistic = LogisticRegression(max_iter=1000)
logistic.fit(X_train, y_train)
```

```
y_pred = logistic.predict(X_test)
print("\nLogistic Regression:")
print("Coefficients:", dict(zip(features, logistic.coef_[0])))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred, normalize='all
```

```
Logistic Regression:
Coefficients: {'TransactionAmt': np.float64(3.6818556114926283e-06), 'card1_
freq': np.float64(-5.199267651708412e-06), 'card1_TransactionAmt_mean': np.f
loat64(-0.0004382491913412886), 'card1_TransactionAmt_dev': np.float64(0.000
43960126755080884), 'addr1_freq': np.float64(-1.6151762784906556e-05)}
Confusion Matrix:
 [[9.64075253e-01 8.46682697e-06]
 [3.59162800e-02 0.00000000e+00]]
```

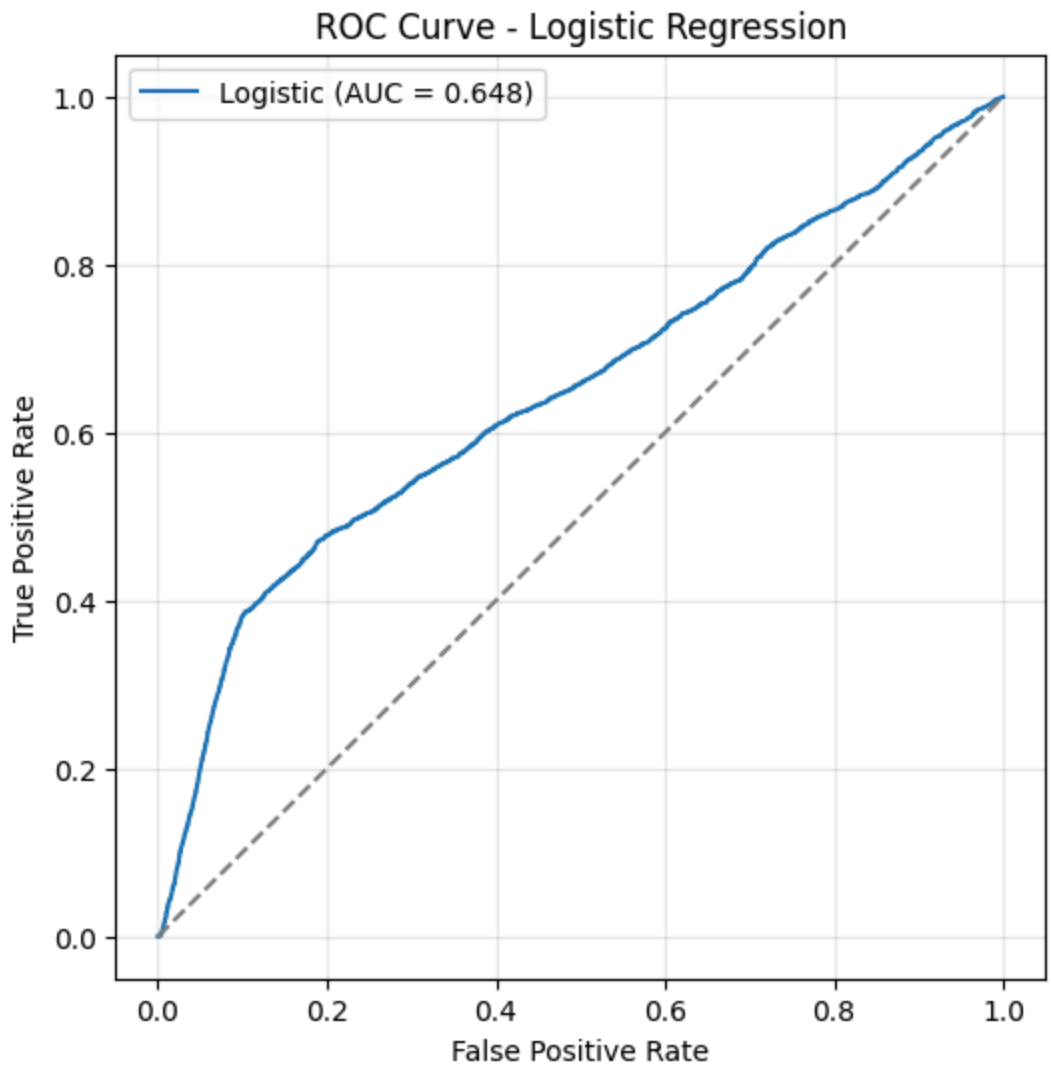
ROC Curve for Logistic regression

AUC = 0.648

This means that the model is barely better than a model that classifies a data point as fraudulent on random but not good enough.

In [49]:
```
y_prob_log = logistic.predict_proba(X_test)[:, 1]

auc_log = roc_auc_score(y_test, y_prob_log)
print("Logistic ROC-AUC:", auc_log)

fpr_log, tpr_log, _ = roc_curve(y_test, y_prob_log)

plt.figure(figsize=(6,6))
plt.plot(fpr_log, tpr_log, label=f'Logistic (AUC = {auc_log:.3f})')
plt.plot([0,1], [0,1], linestyle='--', color='gray')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve - Logistic Regression')
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```

```
Logistic ROC-AUC: 0.6484575970764183
```



XGBoost model

The current best model in the same kaggle competition

In [ ]:
```
from xgboost import XGBClassifier
```

In [ ]:
```
features = [
    'TransactionAmt',
    'card1_freq',
    'card1_TransactionAmt_mean',
    'card1_TransactionAmt_dev',
    'addr1_freq'
]

X_xgb = X[features].fillna(0)
y_xgb = y

X_train, X_test, y_train, y_test = train_test_split(
    X_xgb, y_xgb, test_size=0.2, random_state=42
)
```

```
xgb = XGBClassifier(
    n_estimators=300,
    max_depth=6,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    scale_pos_weight=(len(y_train) / sum(y_train)),  # handle imbalance
    eval_metric='logloss',
    random_state=42
)

xgb.fit(X_train, y_train)
```

Out[ ]:

```
▾ XGBClassifier
▸ Parameters
```

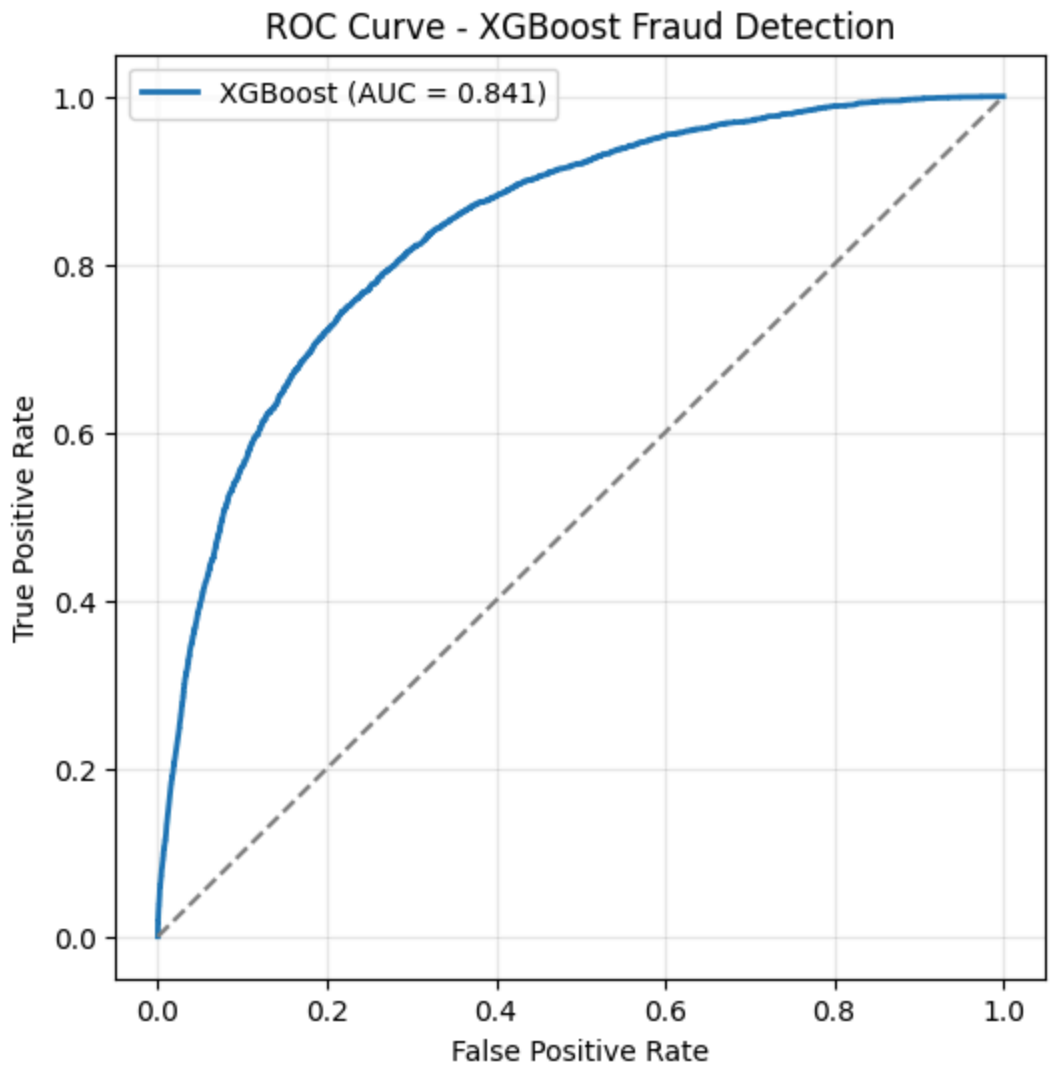
AUC = 0.841

Not bad, starting to get much better. This model blows past the baseline logistic regression model. It does have a high recall of 0.79
Essentially a trade off b/w precision and recall. Since we want to have better fraud detection, we must opt for a higher recall as we cannot let fraud go undetected vs having more false positives.

In [ ]:

```
y_prob = xgb.predict_proba(X_test)[:, 1]
y_pred = xgb.predict(X_test)

print("ROC-AUC:", roc_auc_score(y_test, y_prob))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))

# Compute ROC
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
roc_auc = roc_auc_score(y_test, y_prob)

# Plot
plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f'XGBoost (AUC = {roc_auc:.3f})', linewidth=2)
plt.plot([0,1], [0,1], linestyle='--', color='gray')  # random baseline
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve - XGBoost Fraud Detection')
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```

```
ROC-AUC: 0.8413767299682011
Confusion Matrix:
 [[90001 23865]
 [ 1141  3101]]
              precision    recall  f1-score   support

         0.0       0.99      0.79      0.88    113866
         1.0       0.11      0.73      0.20      4242

    accuracy                           0.79    118108
   macro avg       0.55      0.76      0.54    118108
weighted avg       0.96      0.79      0.85    118108
```



In [55]:
```
v_features = [col for col in X.columns if col.startswith('v')]
print(v_features)
```

```
['v1_sum', 'v1_mean', 'v1_std', 'v1_nan_count', 'v2_sum', 'v2_mean', 'v2_st
d', 'v2_nan_count', 'v3_sum', 'v3_mean', 'v3_std', 'v3_nan_count', 'v4_sum',
'v4_mean', 'v4_std', 'v4_nan_count', 'v5_sum', 'v5_mean', 'v5_std', 'v5_nan_
count', 'v6_sum', 'v6_mean', 'v6_std', 'v6_nan_count', 'v7_sum', 'v7_mean',
'v7_std', 'v7_nan_count']
```

Using the features from the v column in the dataset

Feature list: `['TransactionAmt', 'card1_freq', 'card1_TransactionAmt_mean', 'card1_TransactionAmt_dev', 'addr1_freq''v1_sum', 'v1_mean', 'v1_std', 'v1_nan_count', 'v2_sum', 'v2_mean', 'v2_std', 'v2_nan_count', 'v3_sum', 'v3_mean', 'v3_std', 'v3_nan_count', 'v4_sum', 'v4_mean', 'v4_std', 'v4_nan_count', 'v5_sum', 'v5_mean', 'v5_std', 'v5_nan_count', 'v6_sum', 'v6_mean', 'v6_std', 'v6_nan_count', 'v7_sum', 'v7_mean', 'v7_std', 'v7_nan_count']`

Since we have a lot more features to work with, I decided to go with the v features because all of these features were already numeric and it isn't too much hassle to process it.

In [52]:
```
v_features = [col for col in X.columns if col.startswith('v')]
features_with_v = features + v_features

X_xgb = X[features_with_v].fillna(0)
y_xgb = y

X_train, X_test, y_train, y_test = train_test_split(
    X_xgb, y_xgb, test_size=0.2, random_state=42
)

xgb = XGBClassifier(
    n_estimators=300,
    max_depth=6,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    scale_pos_weight=(len(y_train) / sum(y_train)),  # handle imbalance
    eval_metric='logloss',
    random_state=42
)

xgb.fit(X_train, y_train)
```

Out[52]:

▾ XGBClassifier ⓘ ?

▸ Parameters

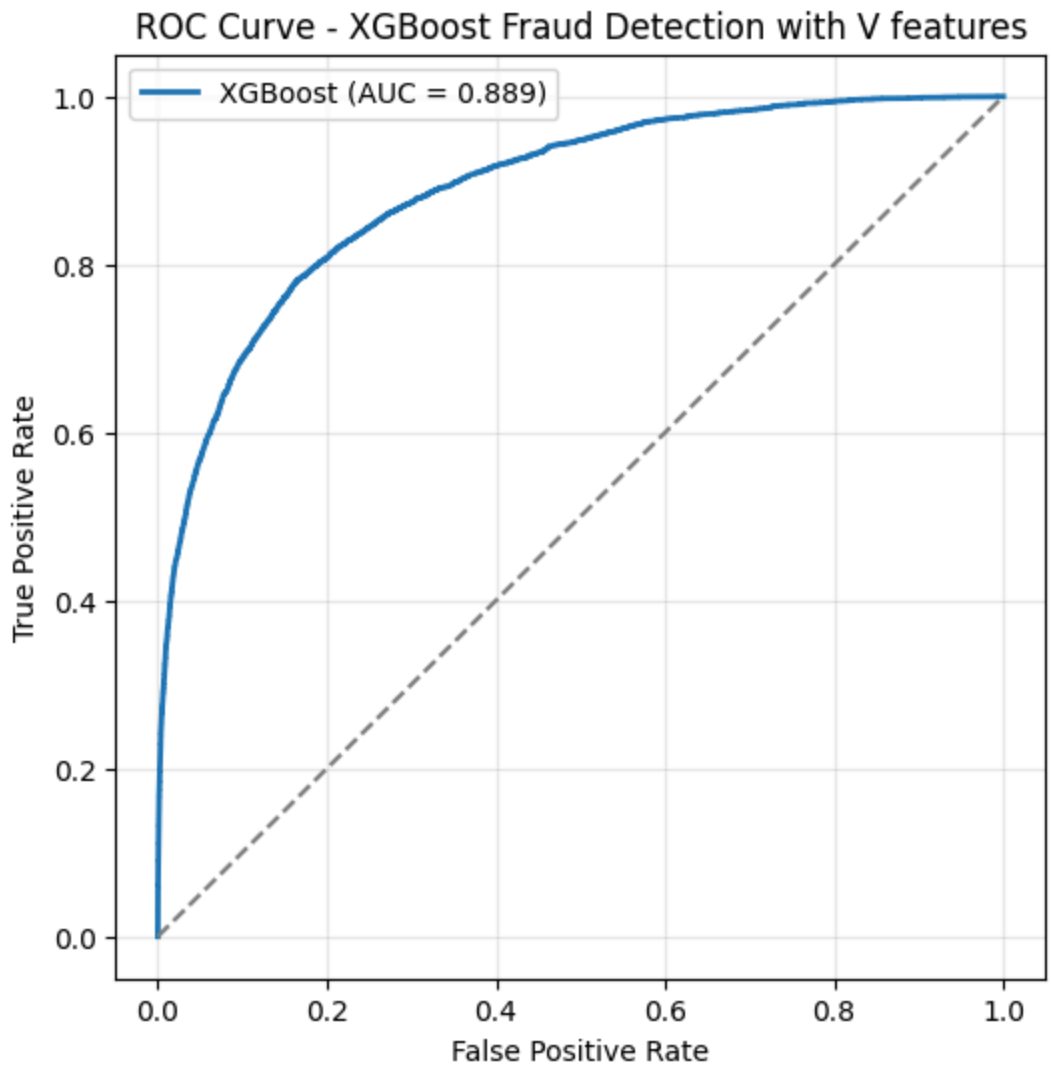
AUC = 0.889

A jump in performance. Recall also has gone higher (0.85), much better than before.

In [54]:
```
y_prob = xgb.predict_proba(X_test)[:, 1]
y_pred = xgb.predict(X_test)
```

```python
print("ROC-AUC:", roc_auc_score(y_test, y_prob))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))

# Compute ROC
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
roc_auc = roc_auc_score(y_test, y_prob)

# Plot
plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f'XGBoost (AUC = {roc_auc:.3f})', linewidth=2)
plt.plot([0,1], [0,1], linestyle='--', color='gray')  # random baseline
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve - XGBoost Fraud Detection with V features')
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```

```
ROC-AUC: 0.8890826871090018
Confusion Matrix:
 [[96553 17313]
 [ 1007  3235]]
              precision    recall  f1-score   support

         0.0       0.99      0.85      0.91    113866
         1.0       0.16      0.76      0.26      4242

    accuracy                           0.84    118108
   macro avg       0.57      0.81      0.59    118108
weighted avg       0.96      0.84      0.89    118108
```


ROC Curve - XGBoost Fraud Detection with V features
XGBoost (AUC = 0.889)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

In [ ]: